

SECOND PUBLIC EXAMINATION

Honour School of Physics Part B: 3 and 4 Year Courses

Honour School of Physics and Philosophy Part B

B3: ATOMIC AND LASER PHYSICS

TRINITY TERM 2021

Monday, 07 June

Opening time: 09:30 am UK Time

**You have two hours writing time to complete the paper
and up to 30 minutes technical time to upload your answer file.**

*Answer **two** questions.*

Write your candidate number, paper name, paper code and a list of the questions that you have answered on the first page, and submit all of your answers as a single PDF file.

Start the answer to each question on a fresh page.

A list of physical constants and conversion factors will be available within Inspira during the examination

You are permitted to use calculators.

The numbers in the margin indicate the weight that the Examiners expect to assign to each part of the question.

1. (a) Consider an atom with magnetic moment given by

$$\boldsymbol{\mu} = -\mu_B \mathbf{L} - g_s \mu_B \mathbf{S},$$

where g_s is the g-factor, μ_B the Bohr magneton, and $\mathbf{L}$ and $\mathbf{S}$ orbital angular momentum and spin operators, respectively. The interaction of this atom with an external magnetic field is described by the Hamiltonian $\mathcal{H}_{ZE} = -\boldsymbol{\mu} \cdot \mathbf{B}$, with $\mathbf{B}$ the external field. Within the LS coupling scheme and considering an hydrogen atom, give a numerical estimate for the magnetic field such that $\mathcal{H}_{ZE}$ can be treated as a perturbation of the fine-structure terms. [4]

(b) Assuming that the magnetic field interaction can be treated as a perturbation, calculate the correction to expectation values of the energy levels and show that this is given by

$$E_{ZE} = g_J \mu_B B M_J,$$

and determine the expression for g_J when $g_s \approx 2$. [5]

(c) Now consider an hydrogen atom, and suppose it is placed in an external potential given by

$$V_{\text{ext}} = \alpha(x^2 + y^2) + \beta z^2,$$

where $0 < \alpha \ll e^2/4\pi\epsilon_0 a_0^3$ and $\beta = -\alpha/2$. Calculate and sketch the five lowest energy levels of the electron in this potential, neglecting any spin-orbit corrections [*Hint: you need to write the total potential in spherical coordinates*]. For which value of β is the ground state energy level not modified by this interaction? [9]

(d) Assume that a weak magnetic field is also applied to the hydrogen atom. Estimate the perturbation to the previously calculated energy levels when $\mathbf{B}$ is parallel to z and parallel to x . [7]

2. (a) Consider an atom with ground state energy E_1 and excited state energy E_2 interacting with a laser beam of frequency $\omega = \omega_0 + \delta$, where $\omega_0 = (E_2 - E_1)/\hbar$, and $|\delta/\omega_0| \ll 1$. The population amplitudes of the lower and upper levels obey the equations:

$$i\frac{dc_1(t)}{dt} = c_2(t)e^{i\delta t}\frac{\Omega}{2},$$

$$i\frac{dc_2(t)}{dt} = c_1(t)e^{-i\delta t}\frac{\Omega}{2}.$$

Discuss the origin and meaning of these equations and explain what is meant by the rotating wave approximation. [4]

(b) Assume that the laser beam is turned on at $t = 0$ and lasts for a short duration τ_p (such that $\omega\tau_p \ll 1$). The laser beam is then turned on again at $t = T$ with the same time duration τ_p . Show that the probability of finding the atom in the upper state is given by:

$$|c_2(t)|^2 = \left|\frac{\Omega\tau_p}{2}\right|^2 \left[\frac{\sin(\delta\tau_p/2)}{\delta\tau_p/2}\right]^2 \cos^2\left(\frac{\delta T}{2}\right). [6]$$

(c) Consider the transition from the $F = 3$, $M_F = 0$ to the $F' = 4$, $M'_F = 0$ levels in the ground state of caesium (its atomic number is $Z = 55$). Assuming that nuclear moment in units of the Bohr magneton is found to be $g_I \approx 4 \times 10^{-4}$, provide an estimate for ω_0 and sketch the upper level probability, $|c_2|^2$, as a function of the laser detuning, δ . [6]

(d) Comment on the similarities between the upper-level transition probability that you have calculated above and what you know from diffraction. [2]

(e) The perturbing laser field also changes the energy of the levels of the atom. When $\Omega = 0$ the coupled atom and photon system has its ground state energy $E'_1 = E_1 + \hbar\omega$ and an excited state energy E_2 . By introducing a new set of variables:

$$\tilde{c}_1 = c_1(t)e^{-i\delta t/2},$$

$$\tilde{c}_2 = c_2(t)e^{i\delta t/2},$$

solve the eigenvalue equations for the two-level system given in part (a) and show that their energy separation changes by $\hbar\Omega^2/2\delta$ when $\Omega \neq 0$ and $|\delta| \gg \Omega$. [7]

3. (a) Explain the meaning of the Einstein A and B coefficients and write down how they are related to each other. Make sure to state their dimensions and physical units. [3]

(b) Consider a laser beam of frequency ν interacting with a gas of two-level atoms. The laser specific intensity, I_ν changes as the laser beam propagates through the medium. Accounting for absorption and emission processes, the equation for I_ν is

$$\frac{dI_\nu}{dz} = -\kappa_\nu I_\nu + j_\nu,$$

where κ_ν is the absorption coefficient (corrected by stimulated emission), j_ν is the emission coefficient, and z the propagation distance. The lower level population density is n_1 with degeneracy g_1 , and the upper level population density is n_2 with degeneracy g_2 . Both the lower and upper level have a spectral lineshape $\phi(\nu)$. If the gas is in thermodynamic equilibrium at a temperature T , show that the quantity $S_\nu = j_\nu/\kappa_\nu$ corresponds to a blackbody at the same temperature, *i.e.*, $S_\nu = B_\nu(T)$. [5]

(c) Assume that both the absorption and the emission coefficients are constant with distance. Show that a solution of the equation for the specific intensity is

$$I_\nu(z) = I_\nu(0)e^{-\kappa_\nu z} + S_\nu(1 - e^{-\kappa_\nu z}).$$
 [4]

(d) Using the results above, consider the case of a laser beam propagating through a medium exhibiting a temperature variation with distance, but locally close to thermal equilibrium at a temperature $T(z)$. Show that after a long propagation distance ($\kappa_\nu z \gg 1$) the laser flux is given by

$$F(z) = \sigma T^4 - \frac{4\sigma T^3}{\pi \kappa_R} \frac{\partial T(z)}{\partial z},$$

where $F(z) = \int I_\nu(z) d\nu$, σ is the Stefan-Boltzmann constant and

$$\frac{1}{\kappa_R} \equiv \frac{\int_0^\infty \frac{1}{\kappa_\nu} \frac{\partial B_\nu(T)}{\partial T} d\nu}{\int_0^\infty \frac{\partial B_\nu(T)}{\partial T} d\nu}.$$
 [7]

(e) Suppose you have a two-level atom in thermal equilibrium with a neutrino field rather than a photon field. Re-derive the relation between the Einstein coefficients and comment on your finding compared to what you know from the case when the atom is in thermal equilibrium with a photon field. [6]

4. (a) The lowest energy levels in the $^{12}_6\text{C}$ carbon atom are found to be at 0 cm^{-1} , 16.4 cm^{-1} , 43.4 cm^{-1} , 10192 cm^{-1} , and 21648 cm^{-1} (the values are given with respect to the ground state energy). Write down their corresponding spectroscopic notation, using term symbols and assuming spin-orbit coupling. [6]

(b) An experiment reveals that an excited state of carbon in the $1s^2 2s^2 2p^3$ configuration decays by emitting photons at $\lambda_1 = 193.09\text{ nm}$ and $\lambda_2 = 247.86\text{ nm}$. Determine the energy level of this excited state. [3]

(c) Now assume that a thin carbon foil is illuminated by either a monochromatic beam of photons with wavelength $\lambda_p = 462.6\text{ nm}$ or a beam of electrons with energy $E_e = 2.68\text{ eV}$. Discuss what you would expect to observe from the carbon sample in both cases. [4]

(d) Consider an atom that in the ground state has mass M , and, when excited to a higher level, has mass $M + \Delta M$. If the transition from the lower to the upper state proceeds with the absorption of a photon, calculate the photon frequency in the laboratory frame when the atom is at rest. [4]

(e) Take a hydrogen-like ^{208}Pb atom, with a nuclear charge $Z = 82$. The nucleus has radius $R = r_0 A^{1/3}$ (with $r_0 = 1.2\text{ fm}$) and the charges (protons) are uniformly distributed inside it. By taking the departure of the Hamiltonian from that of a point nucleus as perturbation, show that the energy level shift in the $1s$ state of ^{208}Pb is

$$\Delta E = \frac{4}{5} Z^2 |E_H| \left(\frac{R}{a_0} \right)^2,$$

where E_H is the ground-state energy level for a hydrogen-like ion and a_0 is the Bohr radius, and provide a numerical estimate for the frequency shift. Discuss qualitatively how this result would change for:

- (i) a $2p$ level in a hydrogen-like Pb ion;
- (ii) a $1s$ level in neutral Pb .

[8]